

1) a) $\vec{p}_i = \vec{p}_f$

$$\vec{p}_i = m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i} + m_3 \vec{v}_{3i}$$

$$= (2, 0, 1) + (-2, 2, 0) + (0, -6, 9) = (0, -4, 10) \text{ kg } \frac{\text{m}}{\text{s}} = \vec{p}_f$$

$$\rightarrow \vec{v}_{1f} = \frac{1}{m_1} (\vec{p}_f \cdot \hat{i}) (\hat{i}) = 0 \text{ m/s}$$

$$\rightarrow \vec{v}_{2f} = \frac{1}{m_2} (\vec{p}_f \cdot (-\hat{j})) (-\hat{j}) = \frac{1}{2} (+4) (-\hat{j}) = -2 \hat{j} \text{ (m/s)}$$

$$\rightarrow \vec{v}_{3f} = \frac{1}{m_3} (\vec{p}_f \cdot \hat{k}) (\hat{k}) = \frac{1}{3} (10) (\hat{k}) = \frac{10}{3} \hat{k} \text{ (m/s)}$$

b) $E_i \stackrel{?}{=} E_f$ no potential $\Rightarrow T_i \stackrel{?}{=} T_f$

$$T_i = \frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 + \frac{1}{2} m_3 v_{3i}^2 = \frac{1}{2} 1 (2^2 + 0^2 + 1^2) + \frac{1}{2} 2 (1^2 + 1^2 + 0^2) + \frac{1}{2} 3 (0^2 + 2^2 + 3^2)$$

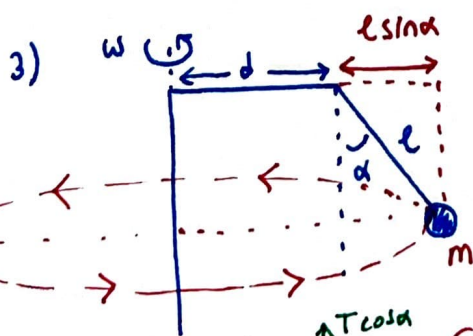
$$= \frac{5}{2} + 2 + \frac{3}{2} 13 = \frac{5}{2} + \frac{4}{2} + \frac{39}{2} = \frac{48}{2} = 24 \text{ J}$$

$$T_f = 0 + \frac{1}{2} 2 (4) + \frac{1}{2} 3 \left(\frac{100}{9}\right) = 4 + \frac{100}{6} = \frac{124}{6} = \frac{62}{3} \text{ J} \quad \frac{62}{3} < 24$$

$T_f < T_i \Rightarrow \text{inelastic}$

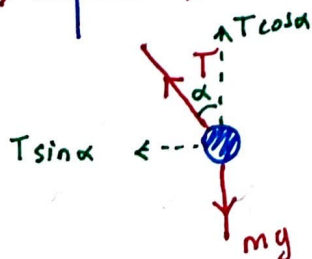
2) The average force applied on the projectile by the wall does a work on it and it reduces the particle's kinetic energy:

$$W = F_{\text{avg}} d = T_f - T_i = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2 = \frac{m}{2} (v_2^2 - v_1^2) \Rightarrow F_{\text{avg}} = \frac{m}{2d} (v_2^2 - v_1^2)$$



radius of circular motion: $d + l \sin \alpha \equiv r$

Eq. of motion: $F = ma = m \frac{v^2}{r}$: centripetal force.
 $= m \frac{r \omega^2}{r} = m r \omega^2$



$$T \cos \alpha = mg$$

$$T \sin \alpha = F = m r \omega^2$$

$$\frac{T \sin \alpha}{T \cos \alpha} = \frac{m r \omega^2}{mg} \Rightarrow \tan \alpha = \frac{r \omega^2}{g} \Rightarrow g \tan \alpha = r \omega^2$$

$$\omega = \sqrt{\frac{g \tan \alpha}{r}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{d + l \sin \alpha}{g \tan \alpha}}$$

$$4) a) F_x = k \cos(\omega t) = m \ddot{x} \Rightarrow \boxed{a = \ddot{x} = \frac{k}{m} \cos(\omega t)} \Rightarrow \dot{x} = \frac{k}{m\omega} \sin(\omega t) + C$$

• Initial cond. $v(0) = 0 \Rightarrow \dot{x}(0) = C = 0 \Rightarrow \boxed{\dot{x}(t) = v(t) = \frac{k}{m\omega} \sin(\omega t)}$

$$x(t) = \int v(t) dt + D = \frac{k}{m\omega} \int \sin(\omega t) dt + D = -\frac{k}{m\omega^2} \cos(\omega t) + D$$

• Ini. cond. $x(0) = 0 \Rightarrow x(0) = -\frac{k}{m\omega^2} + D = 0 \Rightarrow D = \frac{k}{m\omega^2}$

$$\Rightarrow \boxed{x(t) = \frac{k}{m\omega^2} [1 - \cos(\omega t)]}$$

$$b) P(t) = \frac{dW}{dt} = F_x(t) \frac{dx}{dt} = F_x(t) v(t) = k \cos(\omega t) \frac{k}{m\omega} \sin(\omega t)$$

$$\boxed{P(t) = \frac{k^2}{m\omega} \sin(\omega t) \cos(\omega t)}$$